

Table of Contents

2	Continuous Joint Distributions
15	Joint CDFs
20	Independent Continuous Random Variables
30	Bivariate Normal Distribution
37	Sum of Independent Normal Random Variables

16: Continuous Joint Distributions

Jerry Cain

♥ February 14, 2024 ♥

[Lecture Discussion on Ed](#)

A solid orange horizontal bar spanning the width of the slide, positioned at the top.

Continuous Joint Distributions

Stanford logo with darts



The Stanford letterhead logo was created by throwing 500,000 darts according to a joint distribution.

not really, but let's pretend

If we throw another dart according to the same distribution, what is

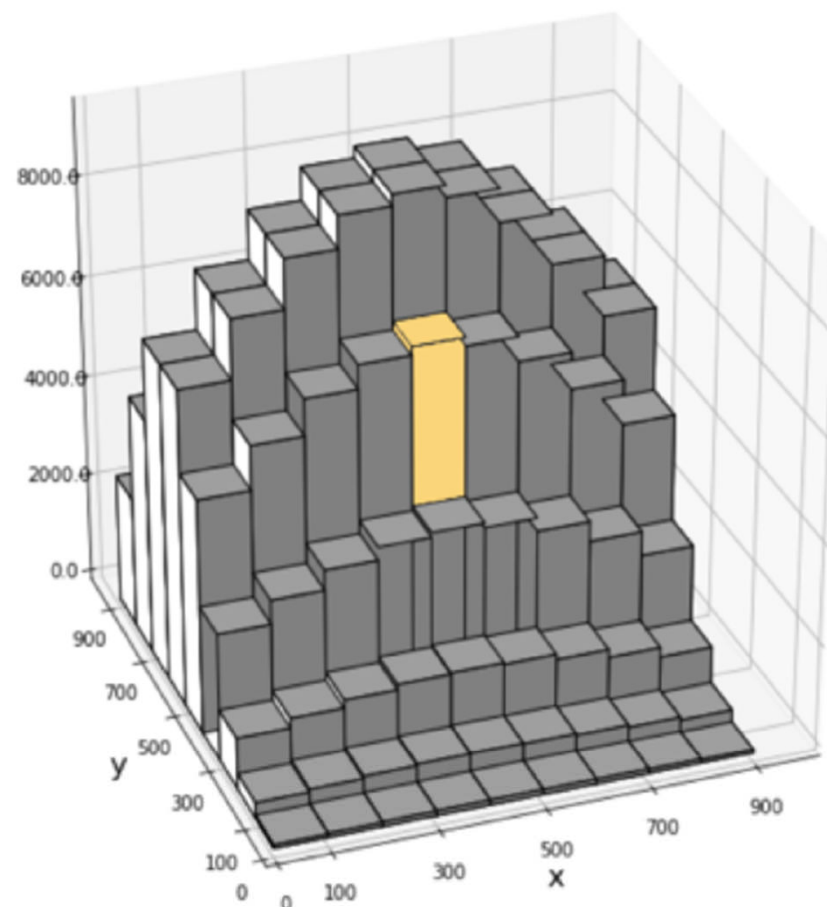
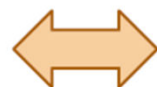
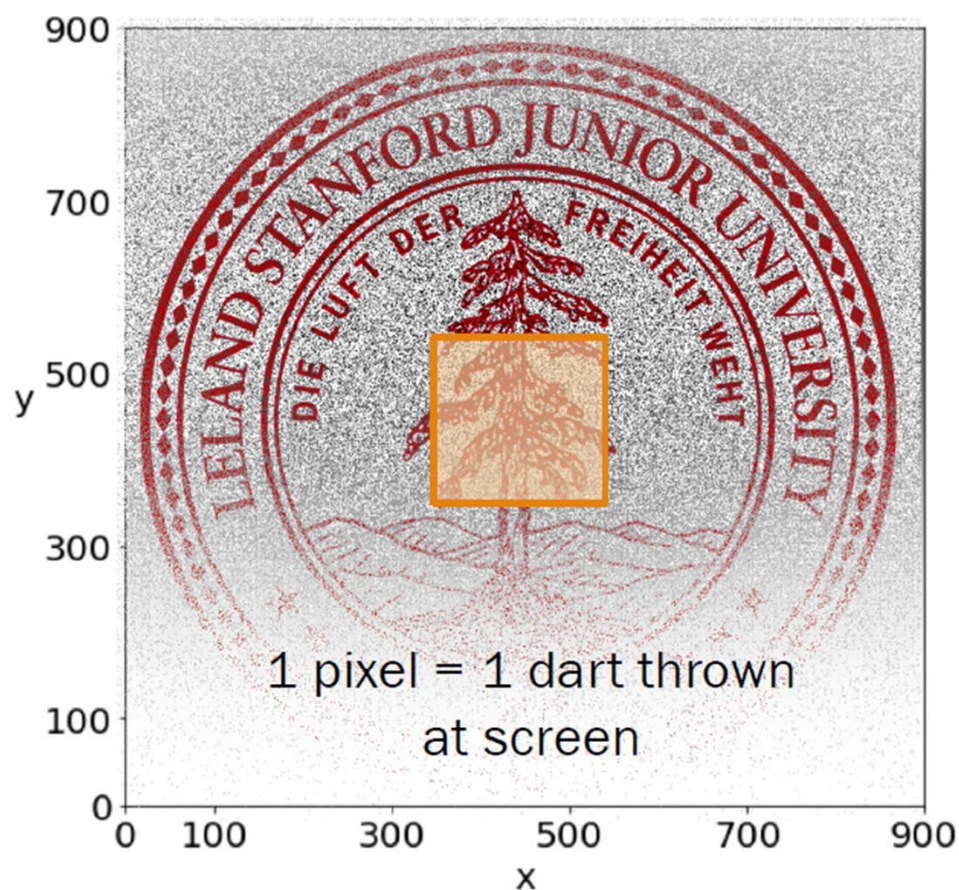
$P(\text{dart hits within } r \text{ pixels of center})?$

Quick check: What is the probability that a dart hits at (456.2344132343, 532.1865739012)?

so small it's essentially zero

CS109 logo with darts

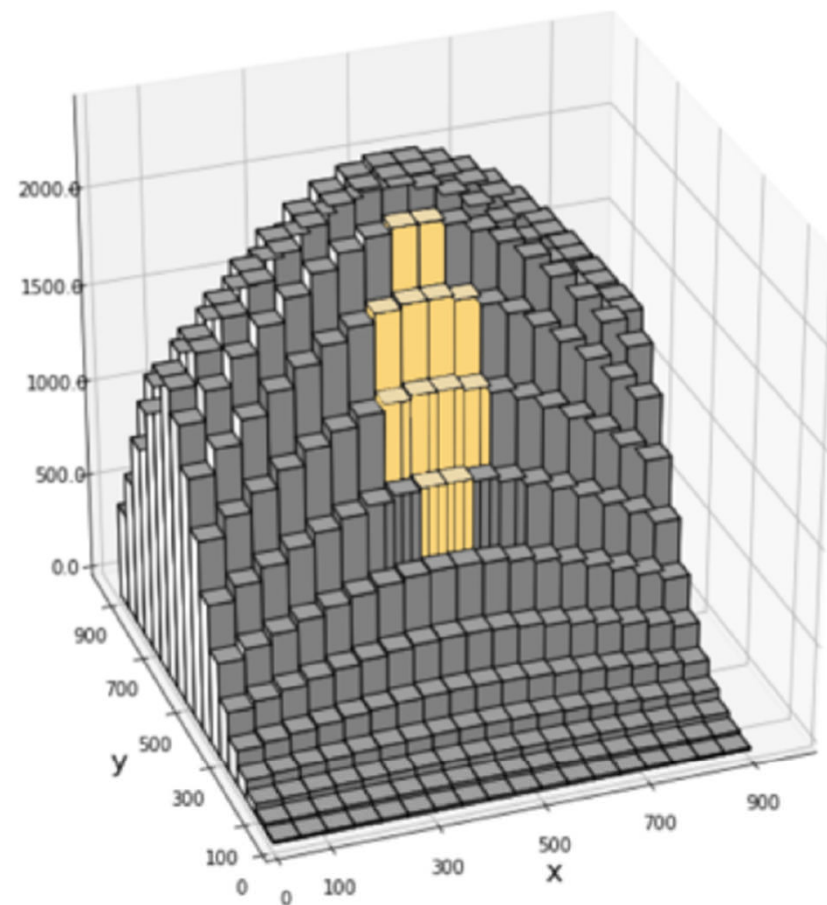
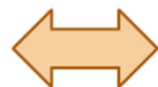
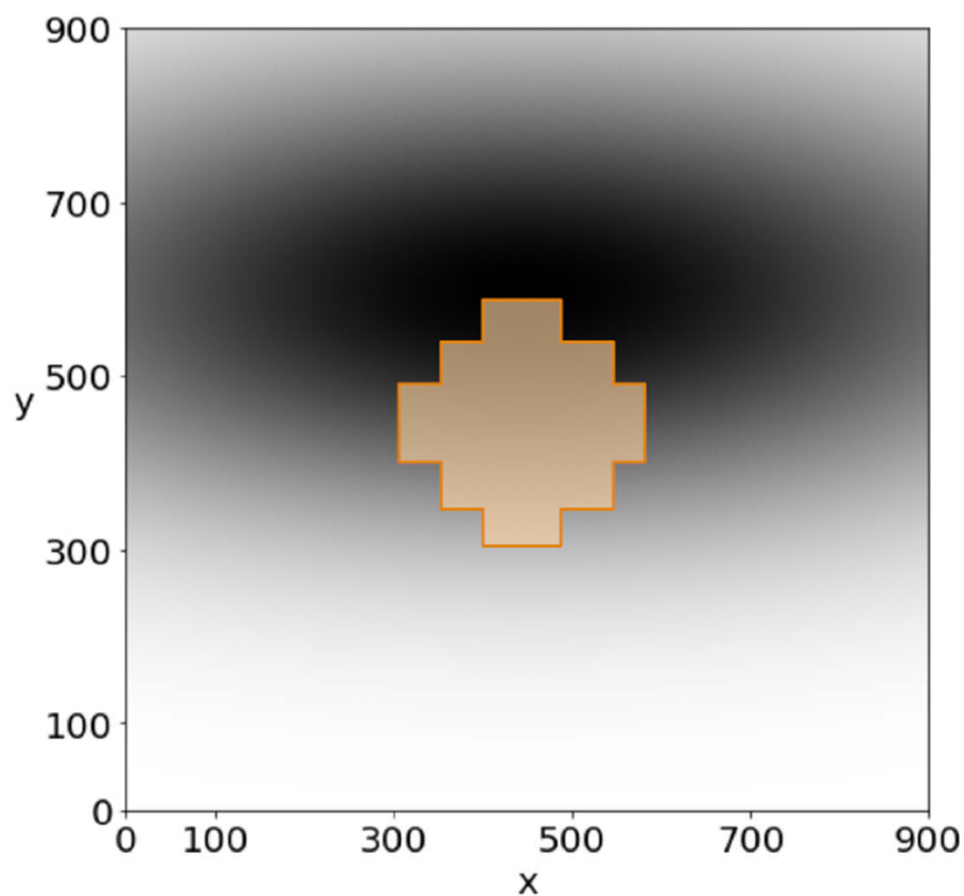
$P(\text{dart hits within } r \text{ pixels of center})?$



Possible dart counts (in 100x100 boxes)

CS109 logo with darts

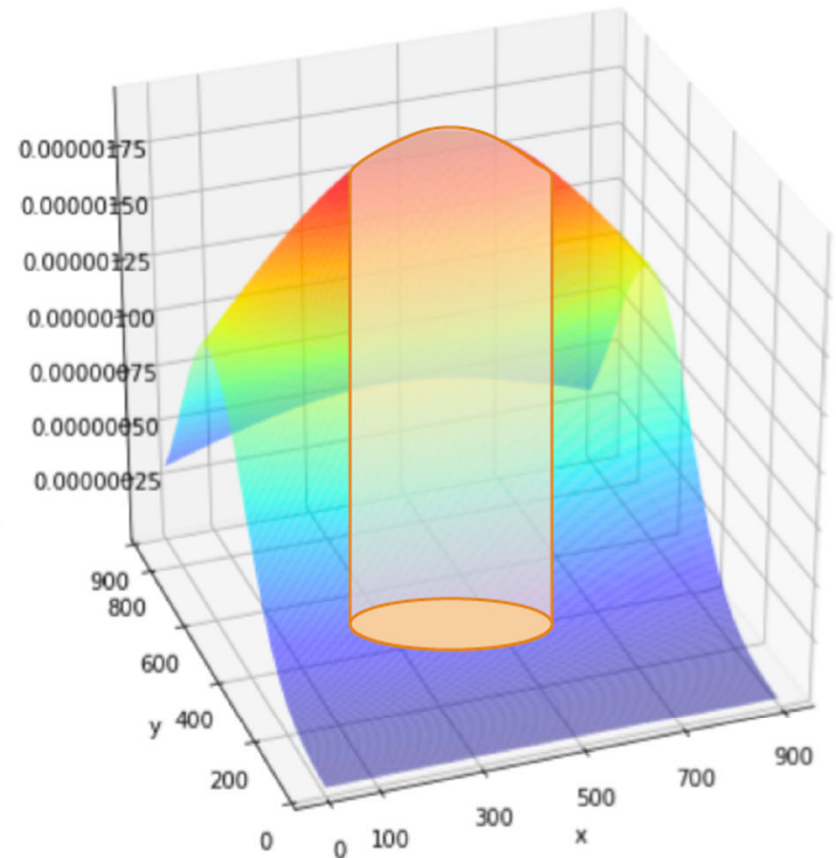
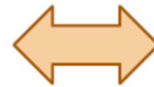
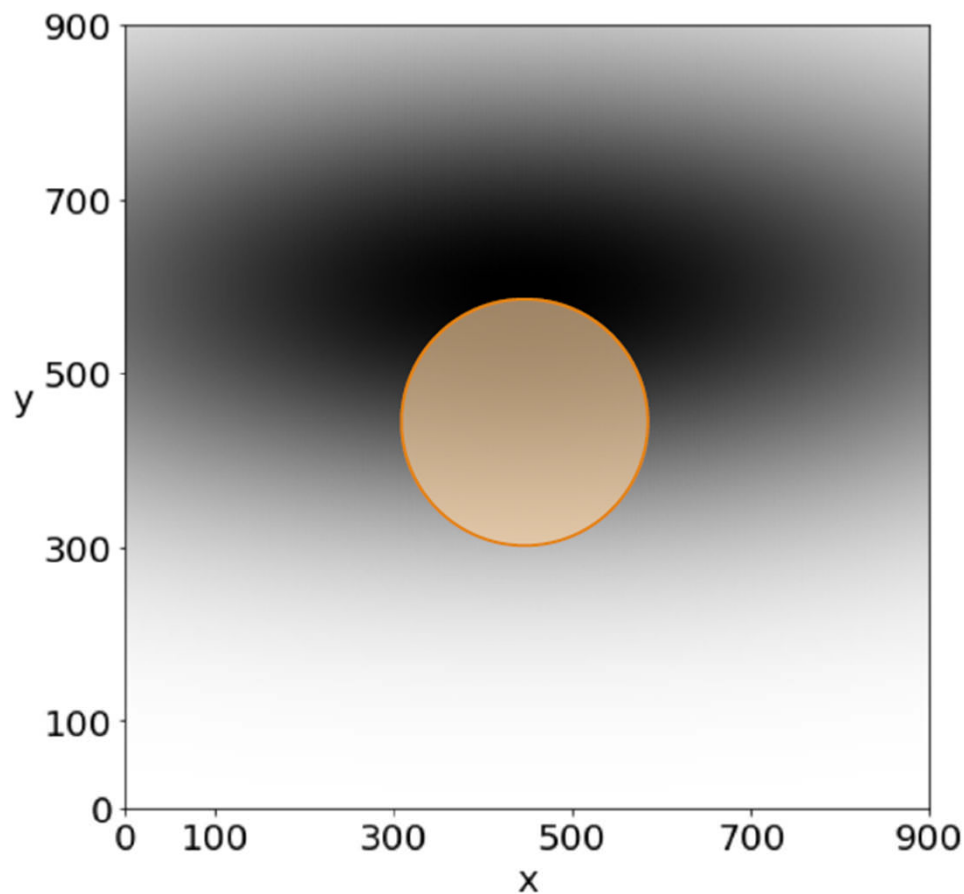
$P(\text{dart hits within } r \text{ pixels of center})?$



Possible dart counts (in 50x50 boxes)

CS109 logo with darts

$P(\text{dart hits within } r \text{ pixels of center})?$



Possible dart counts
(in infinitesimally small boxes) iversity

Continuous joint probability density functions

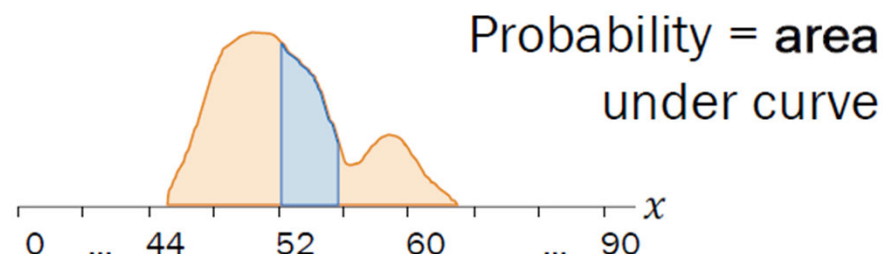
If two random variables X and Y are jointly continuous, then there exists a **joint probability density function** $f_{X,Y}$ defined over $-\infty < x, y < \infty$ such that:

$$P(a_1 \leq X \leq a_2, b_1 \leq Y \leq b_2) = \int_{a_1}^{a_2} \int_{b_1}^{b_2} f_{X,Y}(x, y) dy dx$$

From one continuous RV to jointly continuous RVs

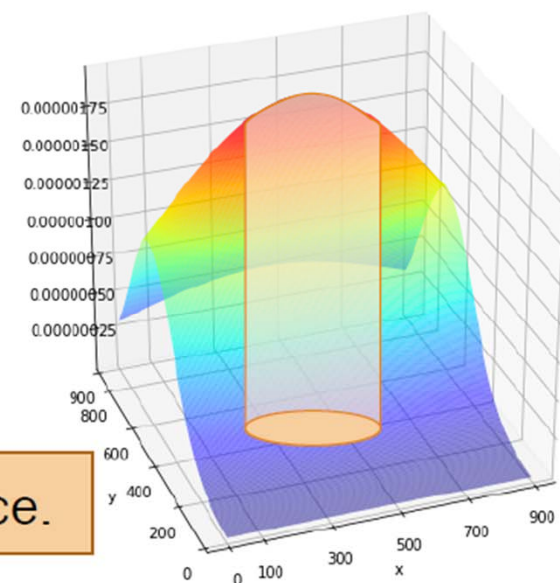
Single continuous RV X

- PDF f_X such that $\int_{-\infty}^{\infty} f_X(x) dx = 1$
- Integrate to get probabilities



Jointly continuous RVs X and Y

- PDF $f_{X,Y}$ such that $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{X,Y}(x,y) dy dx = 1$
- Double integrate to get probabilities



Probability for jointly continuous RVs is **volume** under a surface.

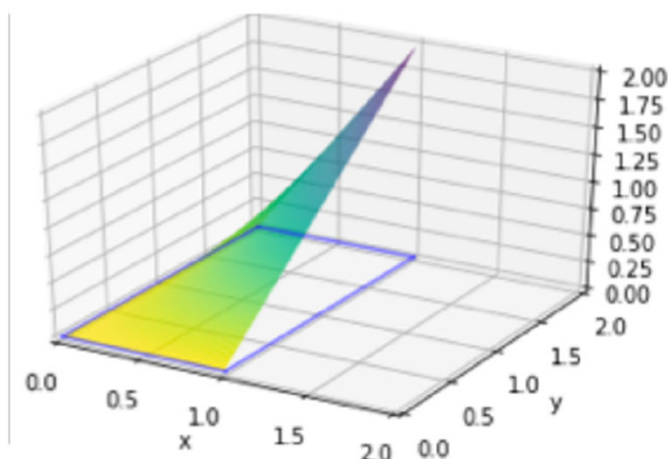
Double integrals without tears

Let X and Y be two continuous random variables.

- Support: $0 \leq X \leq 1$, $0 \leq Y \leq 2$.

Is $g(x, y) = xy$ a valid joint PDF over X and Y ?

Write down the definite double integral that must integrate to 1:



Stanford University

Double integrals without tears

Let X and Y be two continuous random variables.

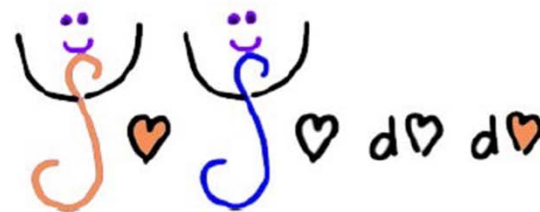
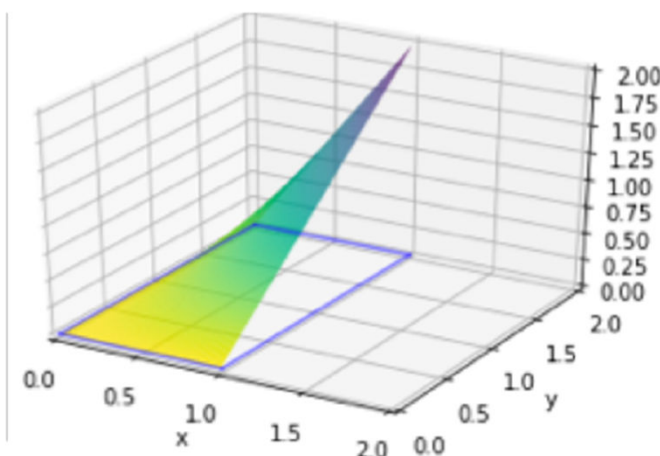
- Support: $0 \leq X \leq 1, 0 \leq Y \leq 2$.

Is $g(x, y) = xy$ a valid joint PDF over X and Y ?

Write down the definite double integral that must integrate to 1:

$$\int_{y=0}^2 \int_{x=0}^1 xy \, dx \, dy = 1 \quad \text{or} \quad \int_{x=0}^1 \int_{y=0}^2 xy \, dy \, dx = 1$$

(used in next slide)



Double integrals without tears

Let X and Y be two continuous random variables.

- Support: $0 \leq X \leq 1, 0 \leq Y \leq 2$.

Is $g(x, y) = xy$ a valid joint PDF over X and Y ?

0. Set up integral:

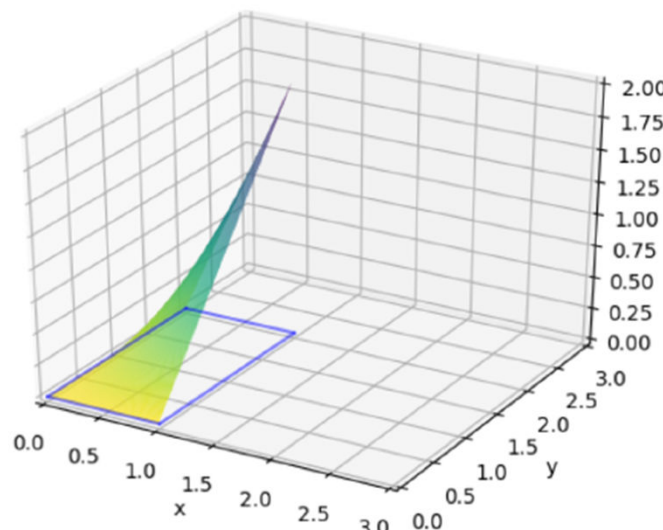
$$1 = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x, y) dx dy = \int_{y=0}^2 \int_{x=0}^1 xy dx dy$$

1. Evaluate inside integral by treating y as a constant:

$$\int_{y=0}^2 \left(\int_{x=0}^1 xy dx \right) dy = \int_{y=0}^2 y \left(\int_{x=0}^1 x dx \right) dy = \int_{y=0}^2 y \left[\frac{x^2}{2} \right]_0^1 dy = \int_{y=0}^2 y \frac{1}{2} dy$$

2. Evaluate remaining (single) integral:

$$\int_{y=0}^2 y \frac{1}{2} dy = \left[\frac{y^2}{4} \right]_{y=0}^2 = 1 - 0 = 1$$

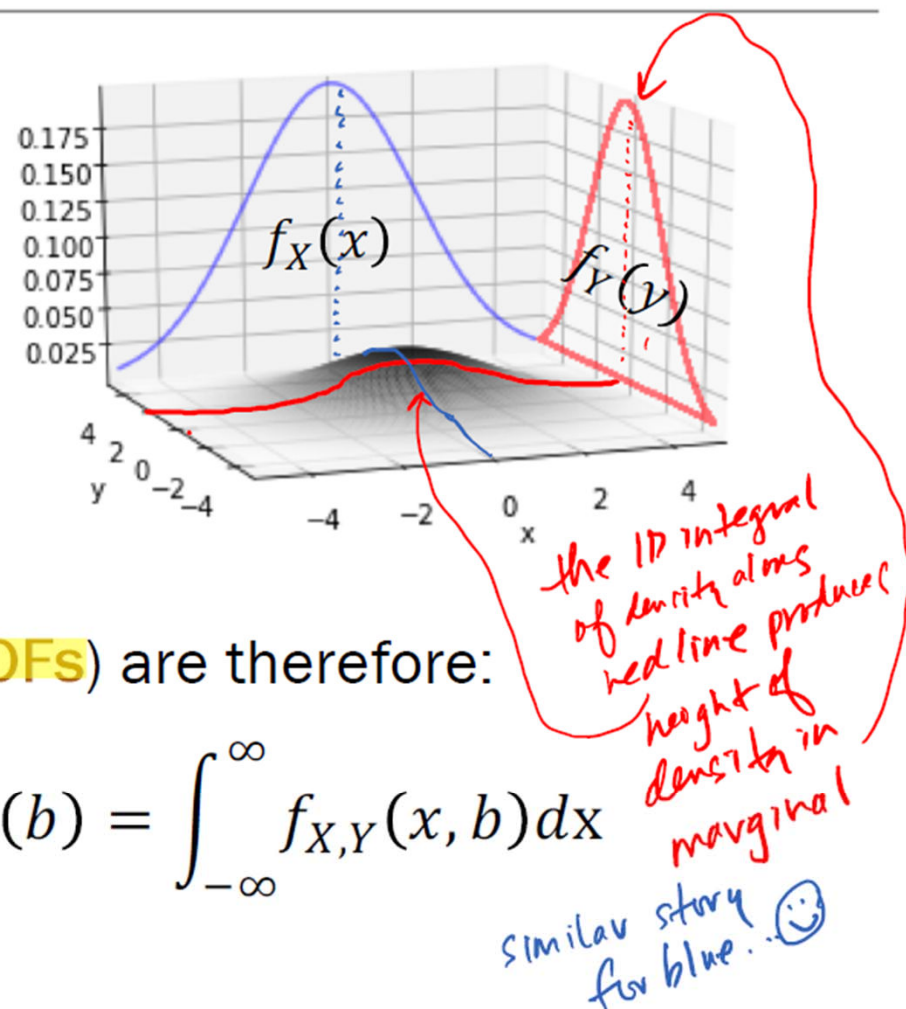


Yes, $g(x, y)$ is a valid joint PDF because it integrates to 1.

Marginal distributions

Suppose X and Y are continuous random variables with joint PDF:

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{X,Y}(x,y) dy dx = 1$$



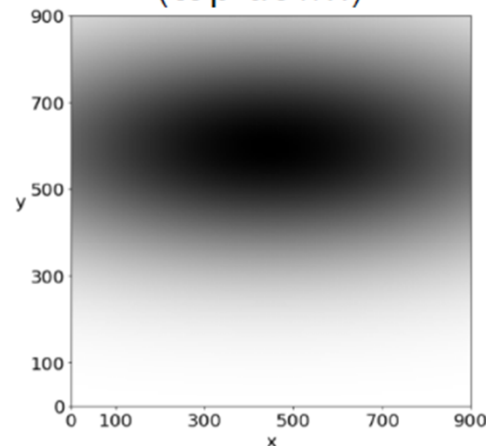
The marginal density functions (**marginal PDFs**) are therefore:

$$f_X(a) = \int_{-\infty}^{\infty} f_{X,Y}(a,y) dy$$

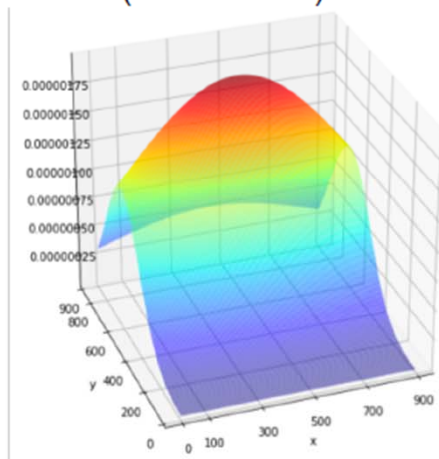
$$f_Y(b) = \int_{-\infty}^{\infty} f_{X,Y}(x,b) dx$$

Back to darts!

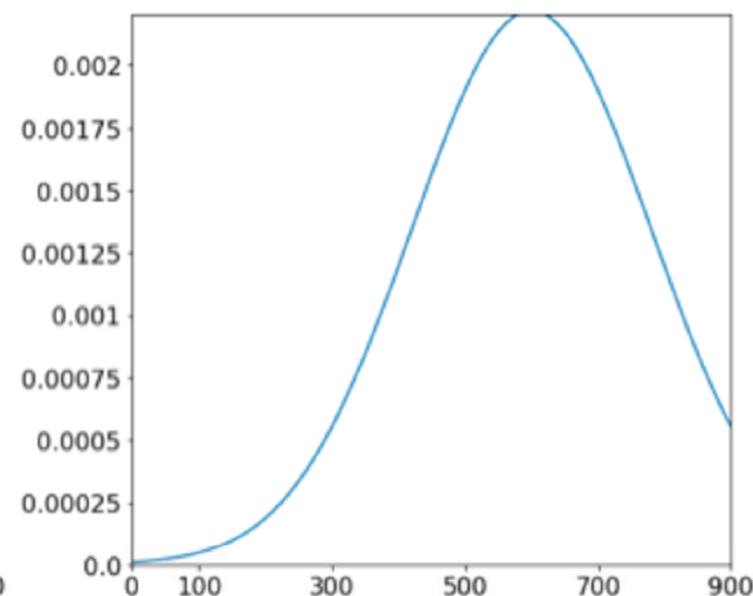
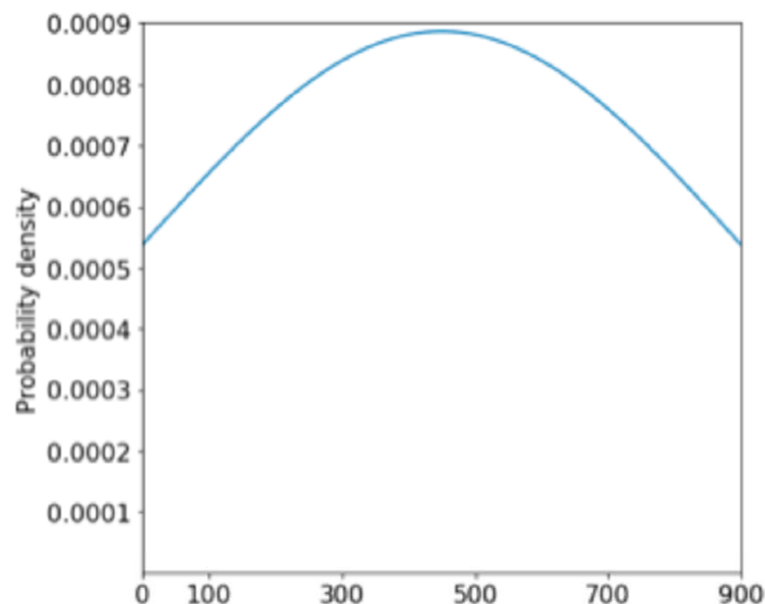
(top-down)



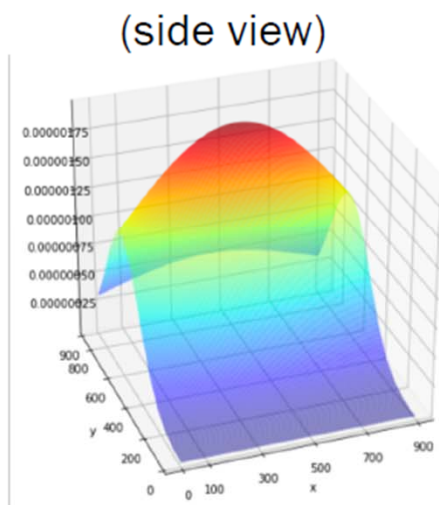
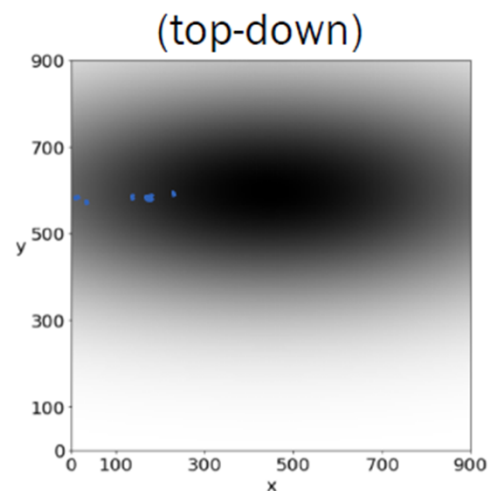
(side view)



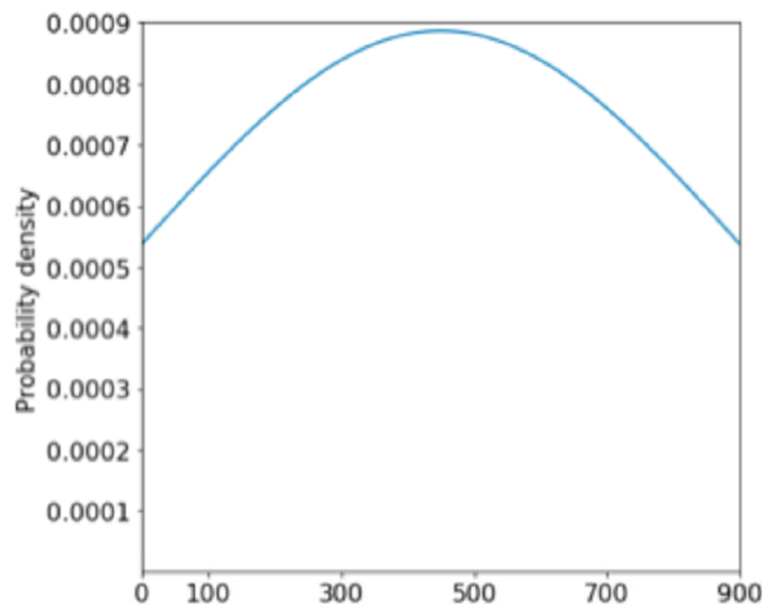
Match X and Y to their respective marginal PDFs:



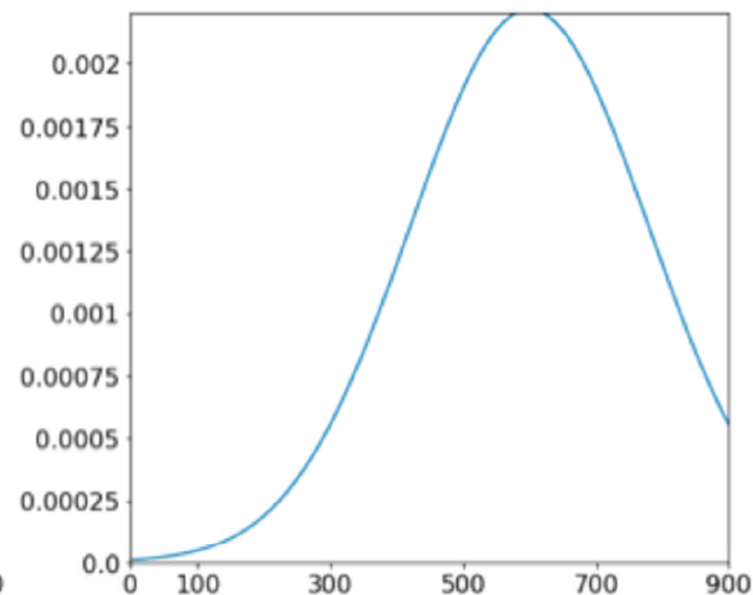
Back to darts!



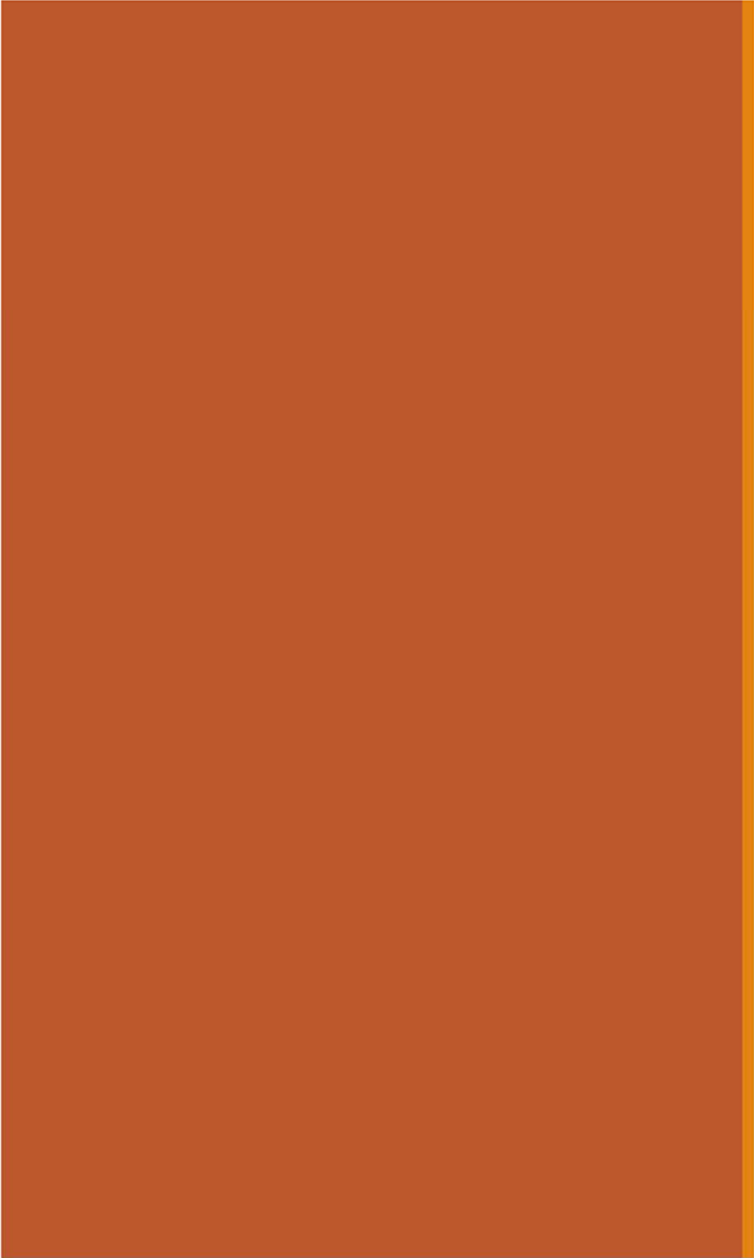
Match X and Y to their respective marginal PDFs:



pixel x



pixel y

A solid orange horizontal bar spanning the width of the slide, positioned near the top.

Joint CDFs

An observation: Connecting CDF to PDF

For a continuous random variable X with PDF f , the CDF (cumulative distribution function) is

$$F(a) = P(X \leq a) = \int_{-\infty}^a f(x)dx$$

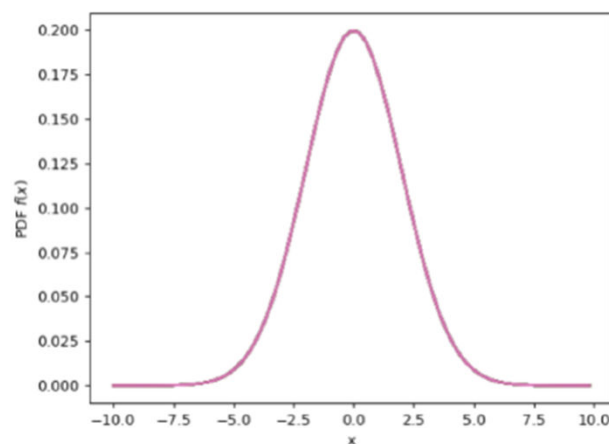
The density f is therefore the derivative of the CDF, F :

$$f(x) = \frac{d}{dx}F(x)$$

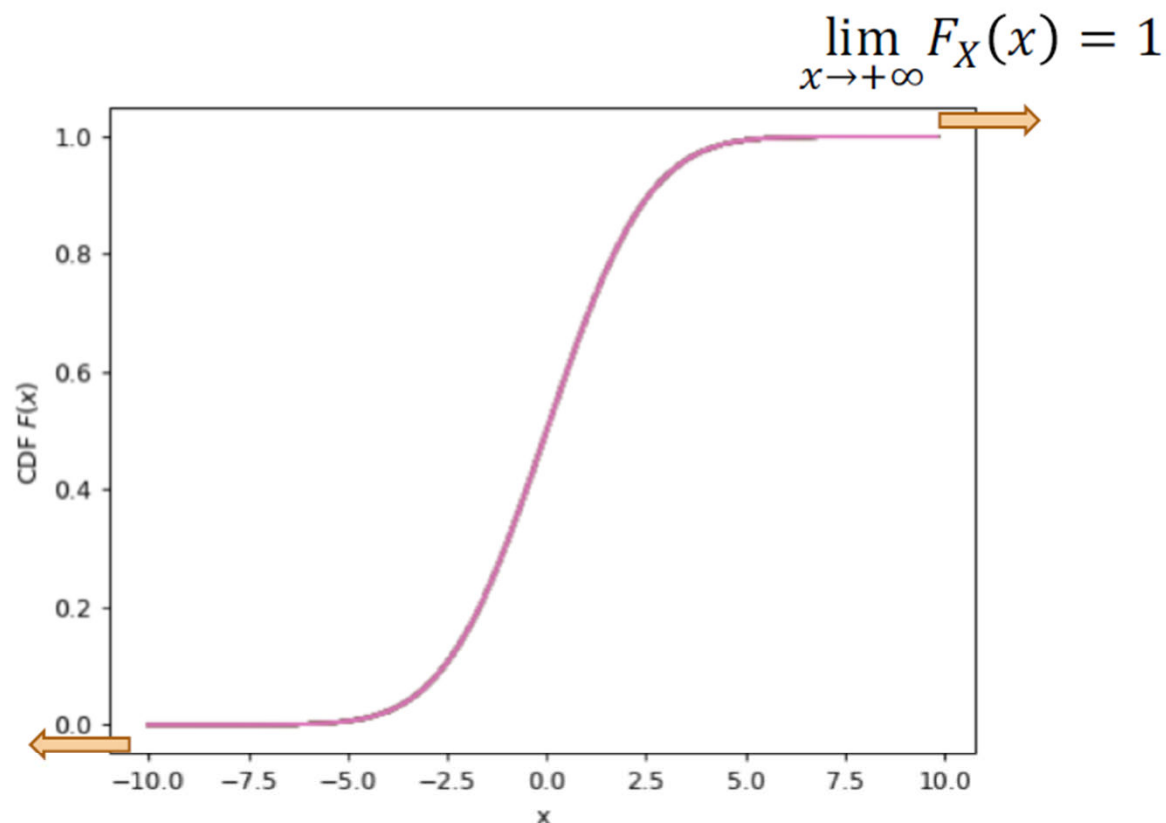
Fundamental Theorem of
Calculus

Single variable CDF, graphically

Review



$f_X(x)$



$$\lim_{x \rightarrow -\infty} F_X(x) = 0$$

$$\lim_{x \rightarrow +\infty} F_X(x) = 1$$

$$F_X(x) = P(X \leq x)$$

Joint cumulative distribution function

For two random variables X and Y , there can be a **joint cumulative distribution function** $F_{X,Y}$:

$$F_{X,Y}(a, b) = P(X \leq a, Y \leq b)$$

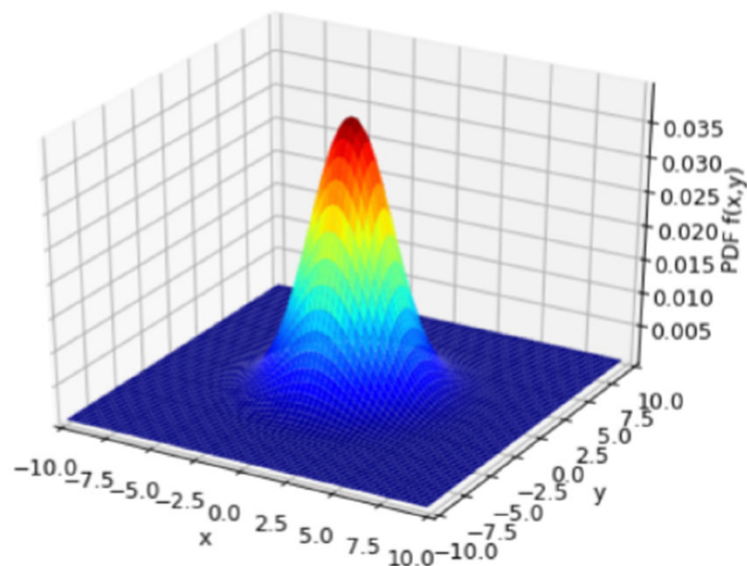
For discrete X and Y :

$$F_{X,Y}(a, b) = \sum_{x \leq a} \sum_{y \leq b} p_{X,Y}(x, y)$$

For continuous X and Y :

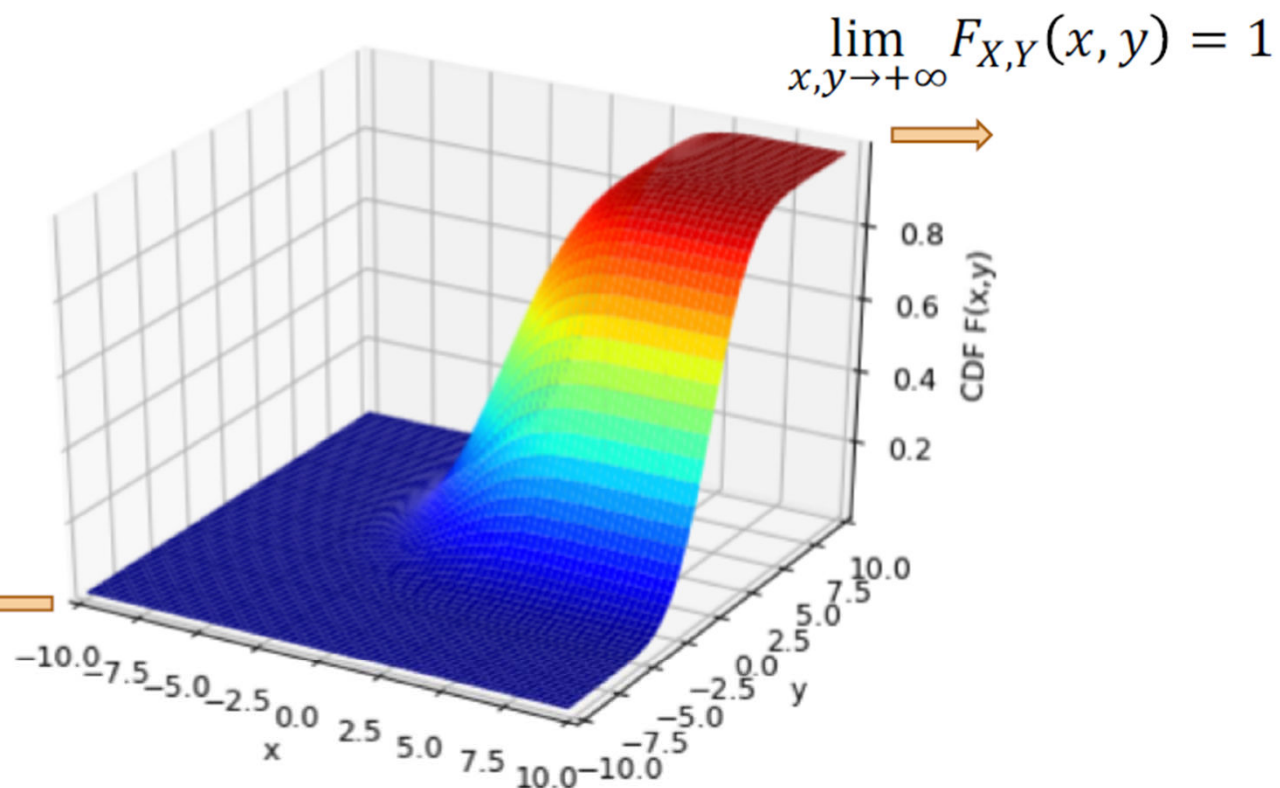
$$F_{X,Y}(a, b) = \int_{-\infty}^a \int_{-\infty}^b f_{X,Y}(x, y) dy dx$$
$$f_{X,Y}(a, b) = \frac{\partial^2}{\partial a \partial b} F_{X,Y}(a, b)$$

Joint CDF, graphically

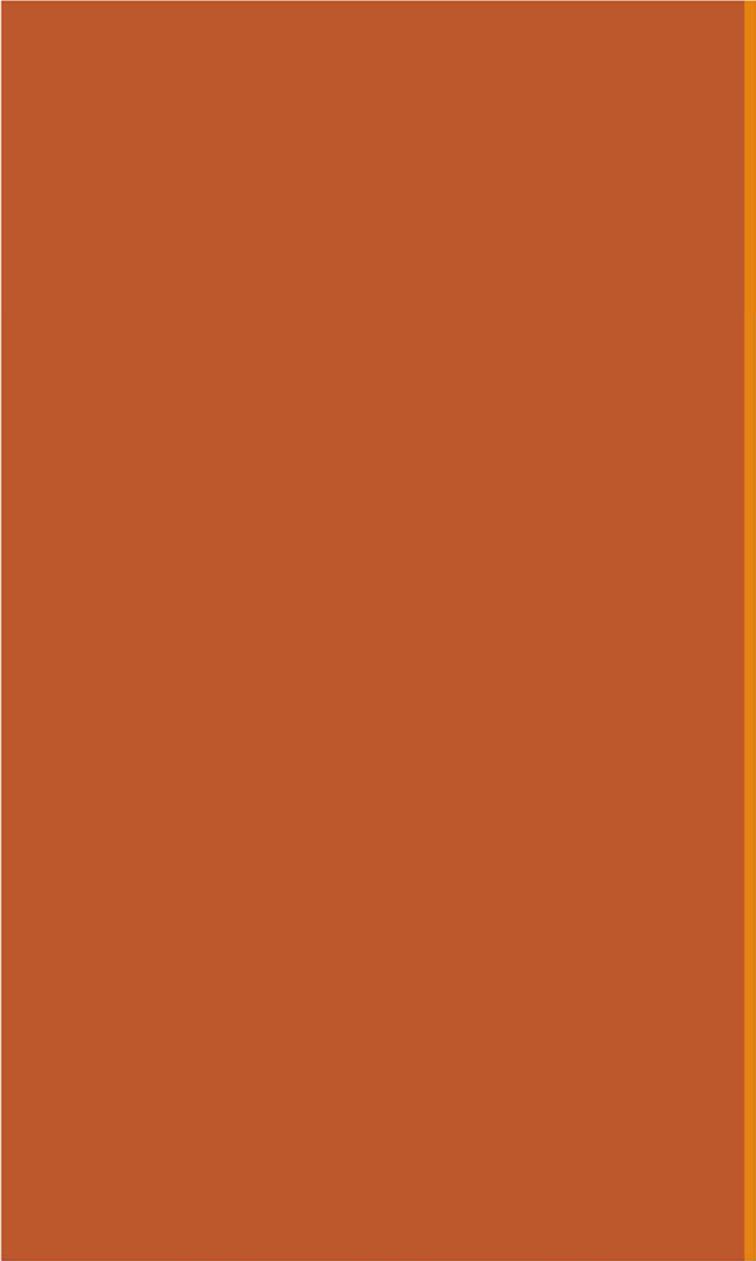


$$\lim_{x,y \rightarrow -\infty} F_{X,Y}(x,y) = 0$$

$$f_{X,Y}(x,y)$$



$$F_{X,Y}(x,y) = P(X \leq x, Y \leq y)$$

A solid orange horizontal bar spanning the width of the slide, positioned at the top.

Independent Continuous RVs

Independent continuous RVs

Two continuous random variables X and Y are **independent** if:

$$P(X \leq x, Y \leq y) = P(X \leq x)P(Y \leq y) \quad \forall x, y$$

Equivalently:

$$\begin{aligned} F_{X,Y}(x, y) &= F_X(x)F_Y(y) \\ f_{X,Y}(x, y) &= f_X(x)f_Y(y) \end{aligned} \quad \forall x, y$$

Proof of PDF:

$$\begin{aligned} f_{X,Y}(x, y) &= \frac{\partial^2}{\partial x \partial y} F_{X,Y}(x, y) = \frac{\partial^2}{\partial x \partial y} F_X(x)F_Y(y) \\ &= \frac{\partial}{\partial x} \frac{\partial}{\partial y} \underline{F_X(x)F_Y(y)} = \frac{\partial}{\partial x} F_X(x) \frac{\partial}{\partial y} F_Y(y) \\ &= f_X(x)f_Y(y) \end{aligned}$$

this is a constant with respect to y!

Independent continuous RVs

Two continuous random variables X and Y are independent if:

$$P(X \leq x, Y \leq y) = P(X \leq x)P(Y \leq y) \quad \forall x, y$$

Equivalently:

$$\begin{aligned} F_{X,Y}(x, y) &= F_X(x)F_Y(y) \\ f_{X,Y}(x, y) &= f_X(x)f_Y(y) \end{aligned} \quad \forall x, y$$

More generally, X and Y are **independent** if joint density on factors into two, single-variable marginal probability densities:

$$f_{X,Y}(x, y) = g(x)h(y), \text{ where } -\infty < \underbrace{x, y}_{\text{states that } x \text{ and } y \text{ can freely roam over their full support!}} < \infty$$

Pop quiz! (more kidding)

$f_{X,Y}(x,y) = g(x)h(y)$,
where $-\infty < x, y < \infty$  independent
X and Y

Are X and Y independent in the following cases?

✓ 1. $f_{X,Y}(x,y) = 6e^{-3x}e^{-2y}$
where $0 < x, y < \infty$

Separable functions: $g(x) = 3e^{-3x}$

$$h(y) = 2e^{-2y}$$

$$\int_0^1 \int_0^1 4xy = 1 = \int_0^1 C 4x dx \cdot \int_0^1 \frac{1}{C} y dy$$

Separable functions: $g(x) = 2x$

$$h(y) = 2y$$

✓ 2. $f_{X,Y}(x,y) = 4xy$
where $0 < x, y < 1$

Constant is
dictated
by requirement
that yellow
integrate to 1 and
blue integrate to 1.
 $C = 1/2$

3. $f_{X,Y}(x,y) = 24xy$
where $0 < x + y < 1$ ✓

Cannot capture constraint on $x + y$!

If you can factor densities over the entire support, you have independence.

More pop quiz! (more kidding)

X and Y have the following joint PDF:

$$f_{X,Y}(x,y) = 3e^{-3x}$$

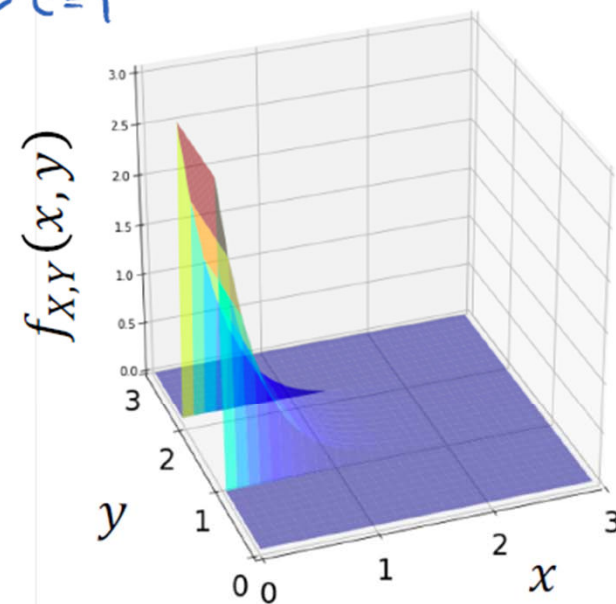
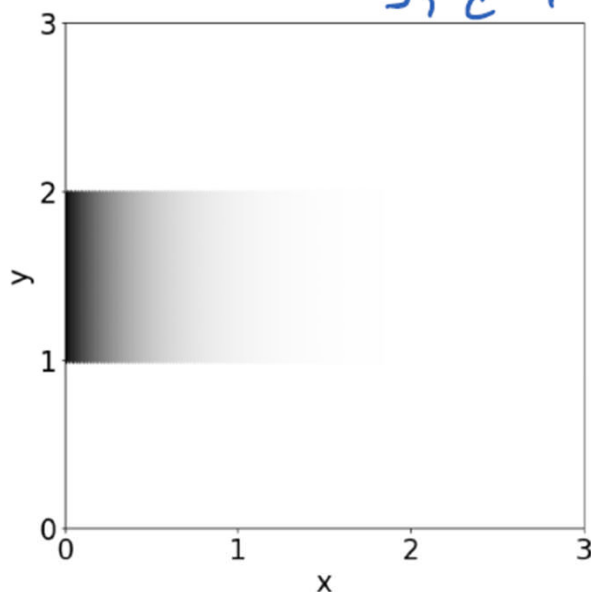
where $0 < x < \infty, 1 < y < 2$

1. Are X and Y independent?

$$g(x) = 3Ce^{-3x}, 0 < x < \infty \quad C \text{ is a}$$
$$h(y) = 1/C, \quad 1 < y < 2 \quad \text{constant}$$
$$\int_1^2 \frac{1}{C} dy = 1 \Rightarrow C = 1$$

2. What is the marginal PDF of X ? Of Y ?

3. What is $E[X + Y]$?



More pop quiz! (more kidding)

X and Y have the following joint PDF:

$$f_{X,Y}(x,y) = 3e^{-3x} \\ \text{where } 0 < x < \infty, 1 < y < 2$$

1. Are X and Y independent?

$$g(x) = 3e^{-3x}, 0 < x < \infty \\ h(y) = 1, 1 < y < 2$$

2. What is the marginal PDF of X ? Of Y ?

$$\begin{cases} f_Y(y) = h(y) = 1, 1 < y < 2 \Leftrightarrow Y \sim \text{Uni}(1,2) \\ f_X(x) = g(x) = 3e^{-3x}, 0 < x < \infty \Leftrightarrow X \sim \text{Exp}(3) \end{cases}$$

3. What is $E[X + Y]$?

$$\text{Strategy 1: } E[g(x,y)] = \int \int g(x,y) f_{X,Y}(x,y) dy dx \quad \text{LOTS} \\ = \int \int (x+y) f_{X,Y}(x,y) dy dx$$

$$\text{Strategy 2: } E[X+Y] = E[X] + E[Y] \\ = 1/3 + 3/2 = 11/6$$

linearity of expectation


The joy of meetings

Two people set up a meeting time. Each arrives independently at a time uniformly distributed between 12pm and 12:30pm.

Define $X = \#$ minutes past 12pm that person 1 arrives. $X \sim \text{Uni}(0, 30)$

$Y = \#$ minutes past 12pm that person 2 arrives. $Y \sim \text{Uni}(0, 30)$

✓ What is the probability that the first to arrive waits > 10 mins for the other?

Compute: $P(X + 10 < Y) + P(Y + 10 < X) = 2P(X + 10 < Y)$ (by symmetry)

1. What is symmetry here?
2. How do we integrate to compute this probability?

$$f_{X,Y}(x,y) = \left(\frac{1}{30}\right)^2$$

$$\underbrace{\iint_{\substack{0 \leq x,y \leq 30 \\ x+10 < y}} \left(\frac{1}{30}\right)^2 dx dy}_{\text{Region 1}} + \underbrace{\iint_{\substack{0 \leq x,y \leq 30 \\ y+10 < x}} \left(\frac{1}{30}\right)^2 dx dy}_{\text{Region 2}}$$

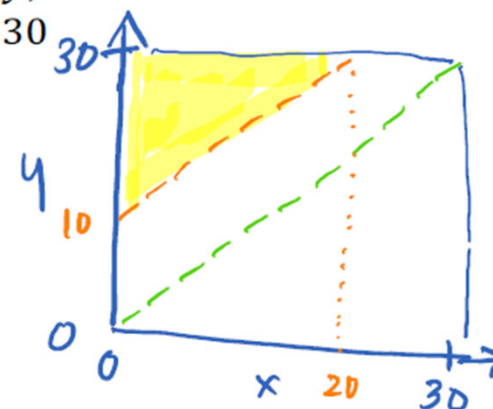
Double integrals: A guide

From last slide: $2P(X + 10 < Y) = 2 \iint_{\substack{x+10 < y, \\ 0 \leq x, y, \leq 30}} (1/30)^2 dx dy$

(by symmetry, independence)

Steps:

1. Draw a picture.
2. Set bounds "from outside in".
 - Outer integral bounds should be full range possible
 - Inner integral can depend on integration variable of outer integral



blue box delineates full range of 2D PDF ✓

green dashed line is $y=x$ ✓

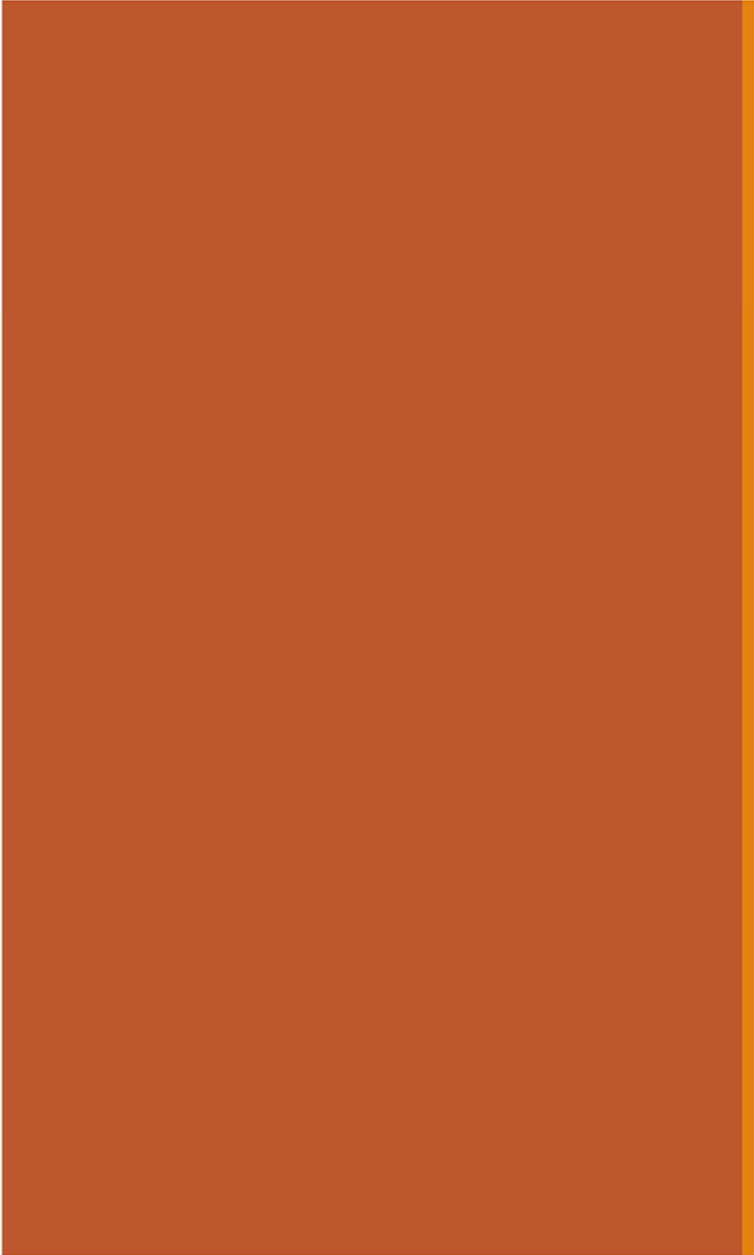
orange dashed line is $y=x+10$ ✓

yellow is everywhere $y > x+10$ ✓

$$= \frac{2}{30^2} \int_{10}^{30} \int_0^{y-10} dx dy$$

$$= \frac{2}{30^2} \int_{10}^{30} (y-10) dy$$

$$= \dots = \frac{9}{9}$$

A solid orange horizontal bar spanning the width of the slide, positioned at the top.

Bivariate Normal Distribution

(랜덤변수 2개)

Bivariate Normal Distribution

recall that for $X \sim \mathcal{N}(\mu, \sigma^2) \Rightarrow f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$

X_1 and X_2 follow a bivariate normal distribution if their joint PDF f is

$$f(x_1, x_2) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} e^{-\frac{1}{2(1-\rho^2)}\left(\frac{(x_1-\mu_1)^2}{\sigma_1^2} - \underbrace{\frac{2\rho(x_1-\mu_1)(x_2-\mu_2)}{\sigma_1\sigma_2}} + \frac{(x_2-\mu_2)^2}{\sigma_2^2}\right)}$$

Can show that $X_1 \sim \mathcal{N}(\mu_1, \sigma_1^2), X_2 \sim \mathcal{N}(\mu_2, \sigma_2^2)$

(Ross chapter 6, example 5d)

Often written as:

$$\underline{\mathbf{X}} \sim \mathcal{N}(\underline{\boldsymbol{\mu}}, \underline{\boldsymbol{\Sigma}})$$

- Vector $\mathbf{X} = (X_1, X_2)$

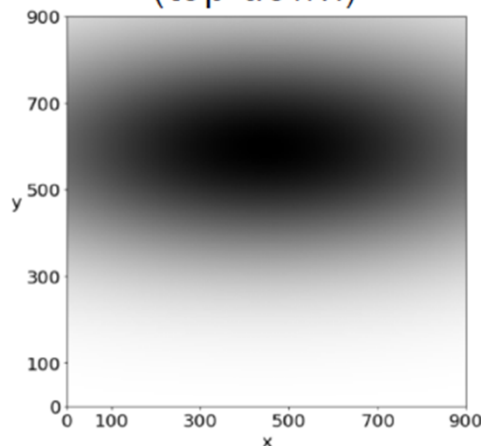
- Mean vector $\boldsymbol{\mu} = (\mu_1, \mu_2)$, Covariance matrix: $\boldsymbol{\Sigma} = \begin{bmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{bmatrix} = \begin{bmatrix} \text{Cov}(X_1, X_1) & \text{Cov}(X_1, X_2) \\ \text{Cov}(X_2, X_1) & \text{Cov}(X_2, X_2) \end{bmatrix}$

Recall correlation: $\rho = \frac{\text{Cov}(X_1, X_2)}{\sigma_1\sigma_2}$

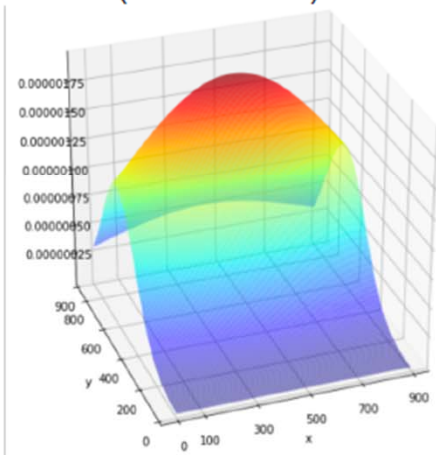
We will focus on understanding the shape of a bivariate Normal RV.

Back to darts

(top-down)



(side view)



Darts were thrown according to a bivariate normal distribution:

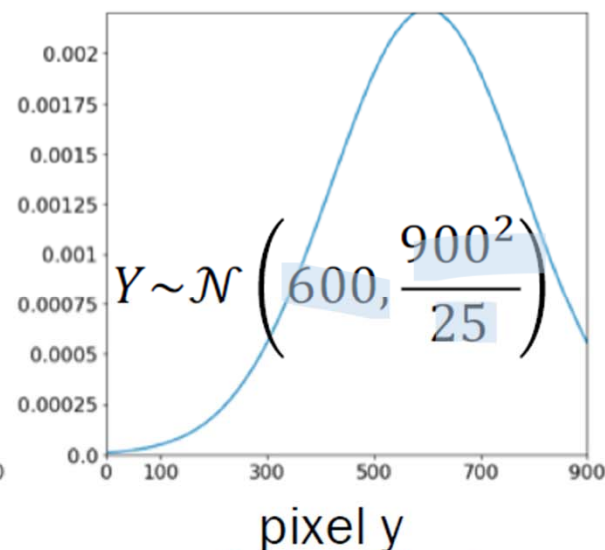
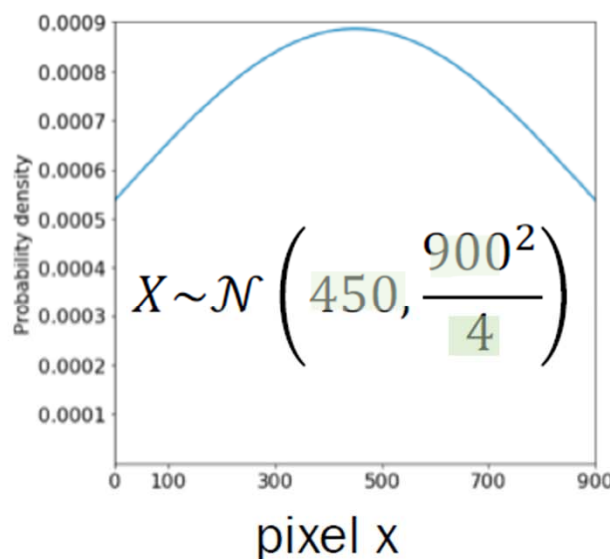
$$(X, Y) \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$$

$$\boldsymbol{\mu} = (450, 600)$$

$$\boldsymbol{\Sigma} = \begin{bmatrix} 900^2/4 & 0 \\ 0 & 900^2/25 \end{bmatrix}$$

says that
 $\text{Cov}(X, Y) = 0$

Marginal
PDFs:



A diagonal covariance matrix

Let $X = (X_1, X_2)$ follow a bivariate normal distribution $X \sim \mathcal{N}(\mu, \Sigma)$, where

$$\mu = (\mu_1, \mu_2),$$

$$\Sigma = \begin{bmatrix} \sigma_1^2 & 0 \\ 0 & \sigma_2^2 \end{bmatrix}$$

still says $\text{Cov}(X_1, X_2) = 0$

Are X_1 and X_2 independent?

$$f(x_1, x_2) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} e^{-\frac{1}{2(1-\rho^2)}\left(\frac{(x_1-\mu_1)^2}{\sigma_1^2} - \underbrace{\frac{2\rho(x_1-\mu_1)(x_2-\mu_2)}{\sigma_1\sigma_2}}_{\text{goes to zero}} + \frac{(x_2-\mu_2)^2}{\sigma_2^2}\right)}$$

$$= \frac{1}{2\pi\sigma_1\sigma_2} e^{-\frac{1}{2}\left(\frac{(x_1-\mu_1)^2}{\sigma_1^2} + \frac{(x_2-\mu_2)^2}{\sigma_2^2}\right)}$$

(Note covariance: $\rho\sigma_1\sigma_2 = 0$)

$$= \frac{1}{\sigma_1\sqrt{2\pi}} e^{-(x_1-\mu_1)^2/2\sigma_1^2} \frac{1}{\sigma_2\sqrt{2\pi}} e^{-(x_2-\mu_2)^2/2\sigma_2^2}$$



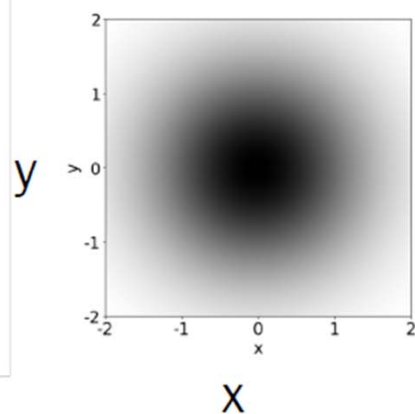
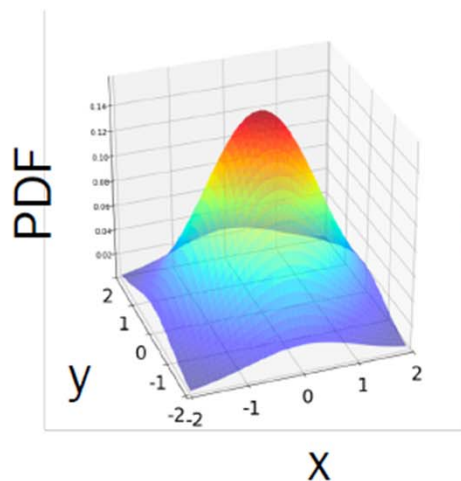
X_1 and X_2 are **independent**
with marginal distributions
 $X_1 \sim \mathcal{N}(\mu_1, \sigma_1^2)$, $X_2 \sim \mathcal{N}(\mu_2, \sigma_2^2)$

(X, Y) Matching (all have $\mu = (0, 0)$)

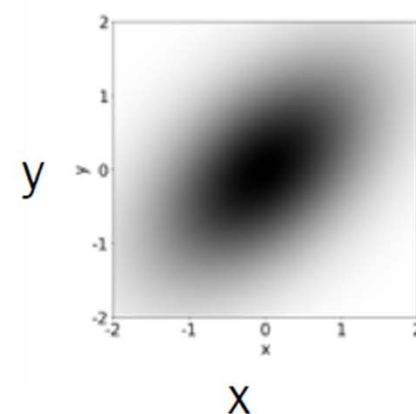
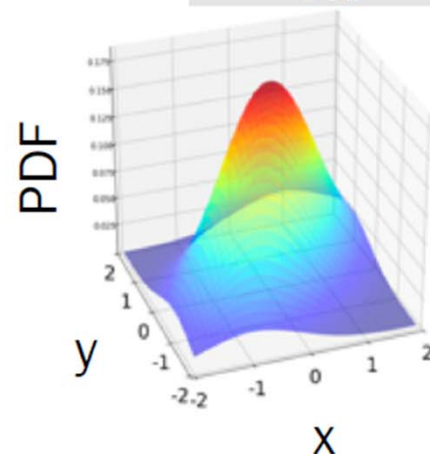


A. $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$
 B. $\begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$
 C. $\begin{bmatrix} 1 & 0.5 \\ 0.5 & 1 \end{bmatrix}$
 D. $\begin{bmatrix} 1 & -0.5 \\ -0.5 & 1 \end{bmatrix}$

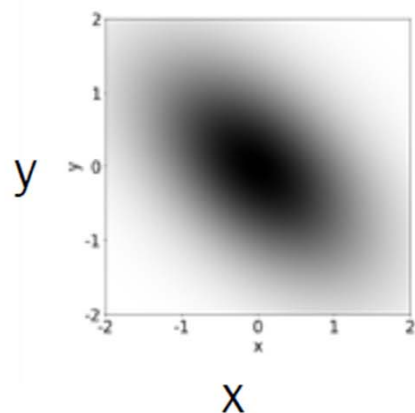
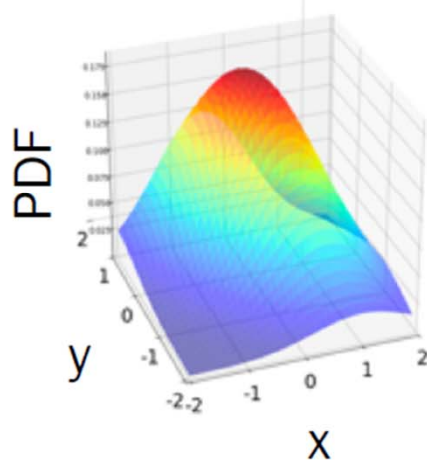
1.



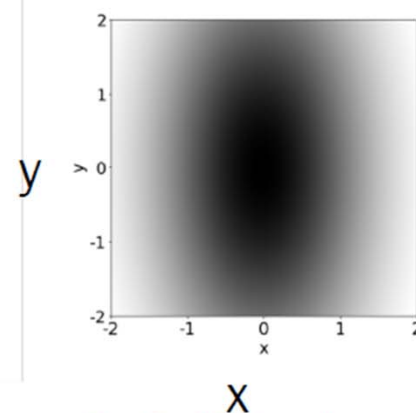
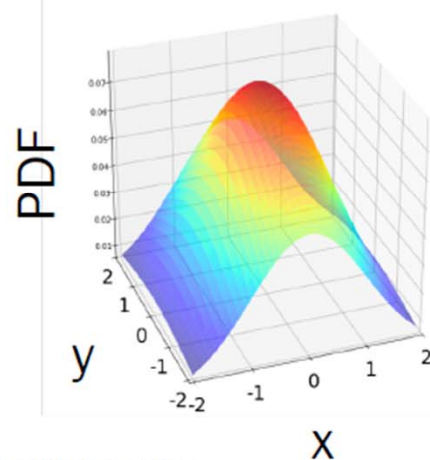
2.



3.



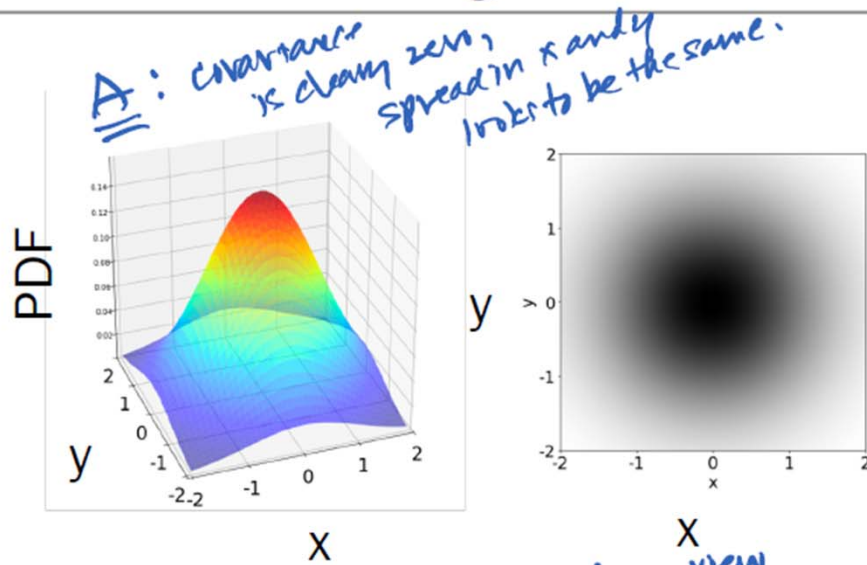
4.



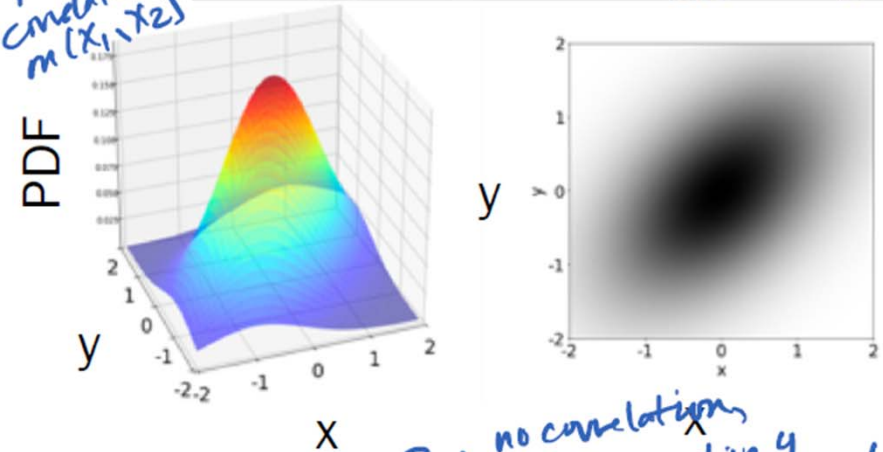
(X, Y) Matching (all have $\mu = (0, 0)$)

A. $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ B. $\begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$
 C. $\begin{bmatrix} 1 & 0.5 \\ 0.5 & 1 \end{bmatrix}$ D. $\begin{bmatrix} 1 & -0.5 \\ -0.5 & 1 \end{bmatrix}$

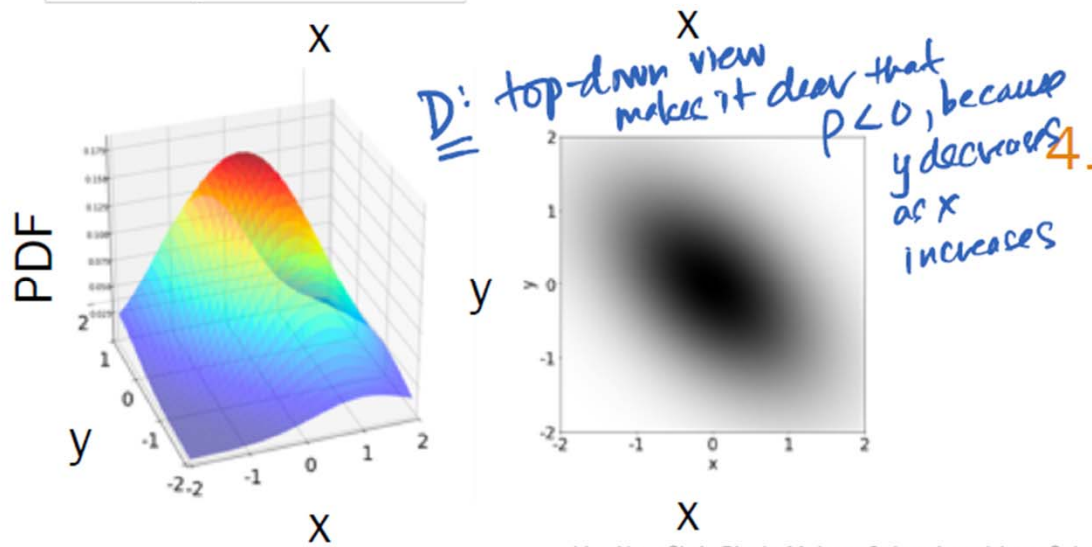
1.



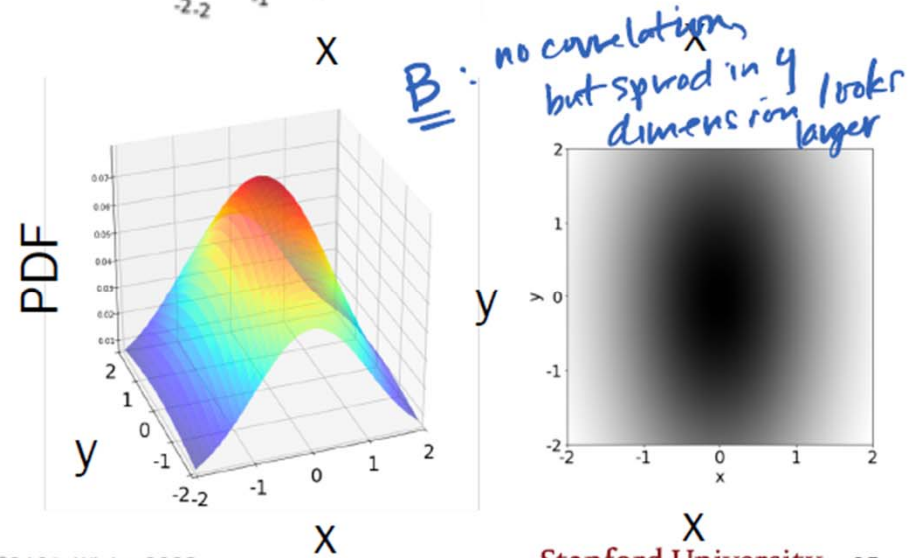
2. C: top-down view suggests positive correlation $\rho(x_1, x_2)$



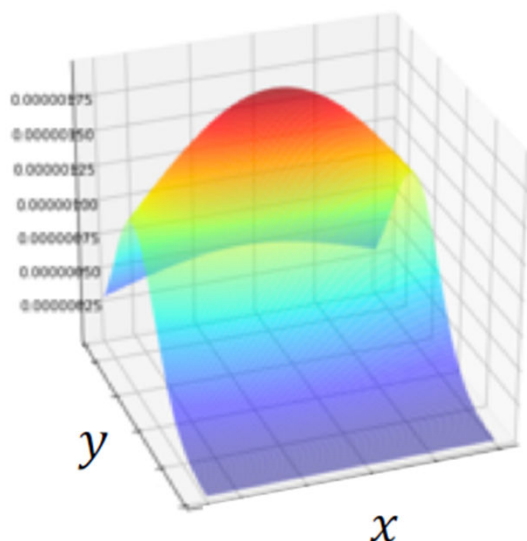
3.



4.



Why are joint PDFs useful?



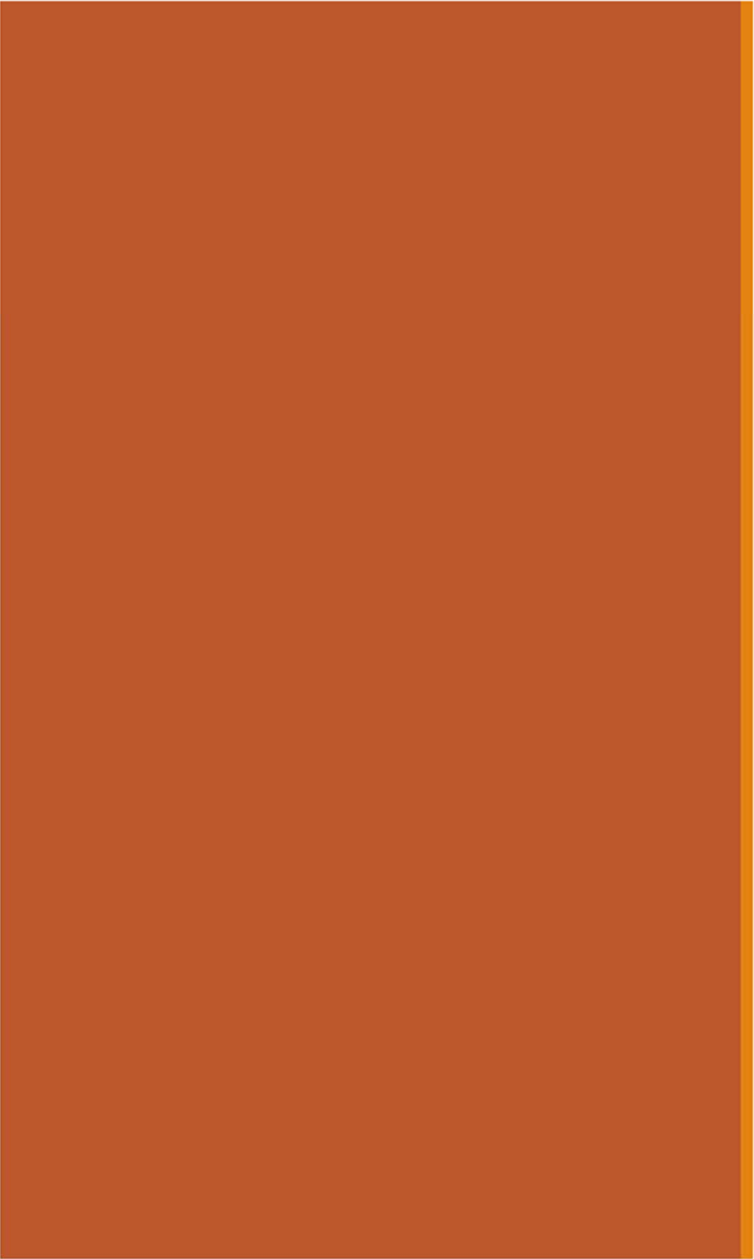
- How 2 continuous RVs vary with each other
- How continuous RV is distributed given evidence (more on Friday)
- How a continuous RV can be decomposed into 2 RVs (or vice versa)

$$P(X < Y), \\ \text{Cov}(X, Y), \rho(X, Y)$$

Given $Y = y$, the distribution of X

Distribution of $Z = X + Y$
(which is a 1-D RV!)

Independence
2-D support
Joint PDF
Joint CDF
Marginal PDF
(Friday) Conditional PDF

A solid orange horizontal bar spanning the width of the slide, positioned at the top.

Sum of Independent Gaussians

Sum of independent Gaussians

$$\Sigma = \begin{bmatrix} \sigma_1^2 & 0 \\ 0 & \sigma_2^2 \end{bmatrix}$$

$$E[X+Y] = E[X] + E[Y] = \mu_1 + \mu_2$$

$$\text{Var}(X+Y) = \text{Var}(X) + \text{Var}(Y) = \sigma_1^2 + \sigma_2^2$$

because of independence

$$\begin{array}{l} X \sim \mathcal{N}(\mu_1, \sigma_1^2), \\ Y \sim \mathcal{N}(\mu_2, \sigma_2^2) \\ X, Y \text{ independent} \end{array} \Rightarrow X + Y \sim \mathcal{N}(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$$

(proof left to [Wikipedia](#))

Holds in general case:

$$\begin{array}{l} X_i \sim \mathcal{N}(\mu_i, \sigma_i^2) \\ X_i \text{ independent for } i = 1, \dots, n \end{array} \Rightarrow \sum_{i=1}^n X_i \sim \mathcal{N}\left(\sum_{i=1}^n \mu_i, \sum_{i=1}^n \sigma_i^2\right)$$

Back for another playoffs game



What is the probability that the Warriors win?
How do you model zero-sum games?

$$P(A_W > A_B)$$

This is a probability of an event involving **two** random variables!

We will compute:

$$P(\underbrace{A_W - A_B}_{\text{A sum of Normals!}} > 0)$$

A sum of Normals!

Motivating idea: Zero sum games



Want: $P(\text{Warriors win}) = P(A_W - A_B > 0)$

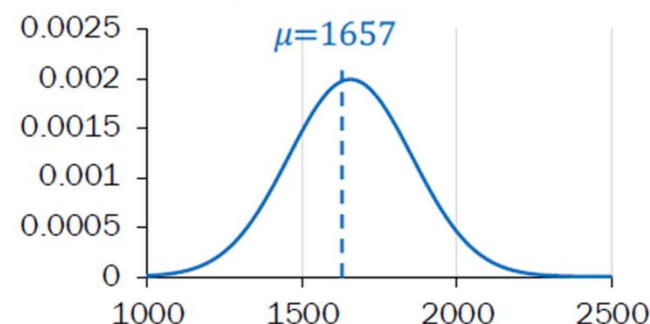
Assume A_W, A_B are independent.

Let $D = A_W - A_B$.

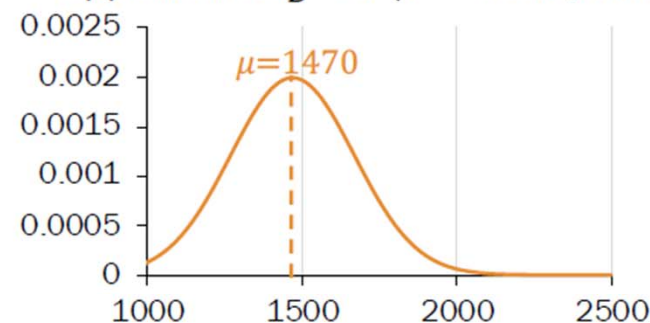
What is the distribution of D ?

- A. $D \sim \mathcal{N}(1657 - 1470, 200^2 - 200^2)$
- B. $D \sim \mathcal{N}(1657 - 1470, 200^2 + 200^2)$
- C. $D \sim \mathcal{N}(1657 + 1470, 200^2 + 200^2)$
- D. $D \sim \mathcal{N}(1657 + 1470, 200^2)$

Warriors $A_W \sim \mathcal{N}(S = 1657, 200^2)$



Opponents $A_B \sim \mathcal{N}(S = 1470, 200^2)$



Motivating idea: Zero sum games



Want: $P(\text{Warriors win}) = P(A_W - A_B > 0)$

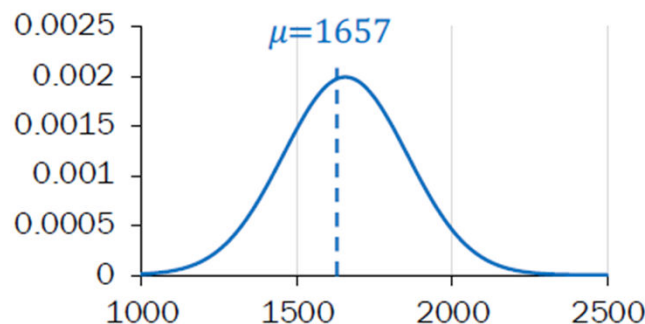
Assume A_W, A_B are independent.

Let $D = A_W - A_B$. } This is really $A_W + (-A_B)$
 $-A_B \sim \mathcal{N}(-1470, (-1)^2 200^2)$
 $\Downarrow$
 $\mathcal{N}(-1470, 200^2)$

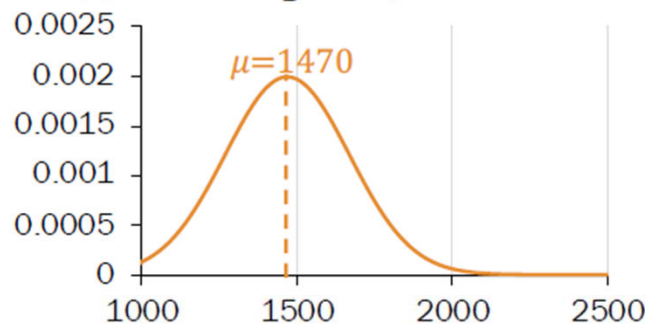
What is the distribution of D ?

- A. $D \sim \mathcal{N}(1657 - 1470, 200^2 - 200^2)$
- B. $D \sim \mathcal{N}(1657 - 1470, 200^2 + 200^2)$**
- C. $D \sim \mathcal{N}(1657 + 1470, 200^2 + 200^2)$
- D. $D \sim \mathcal{N}(1657 + 1470, 200^2)$

Warriors $A_W \sim \mathcal{N}(S = 1657, 200^2)$



Opponents $A_B \sim \mathcal{N}(S = 1470, 200^2)$



If $X \sim \mathcal{N}(\mu_1, \sigma^2)$,
then $(-X) \sim \mathcal{N}(-\mu, (-1)^2 \sigma^2 = \sigma^2)$.

Motivating idea: Zero sum games



Want: $P(\text{Warriors win}) = P(A_W - A_B > 0)$

Assume A_W, A_B are independent.

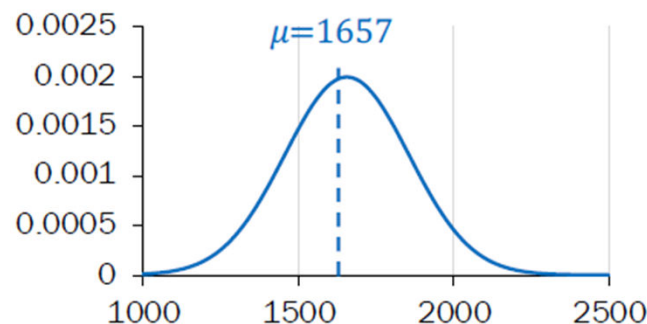
Let $D = A_W - A_B$.

$$\begin{aligned} D &\sim \mathcal{N}(1657 - 1470, 200^2 + 200^2) \\ &\sim \mathcal{N}(187, 2 \cdot 200^2) \quad \sigma \approx 282.842 \end{aligned}$$

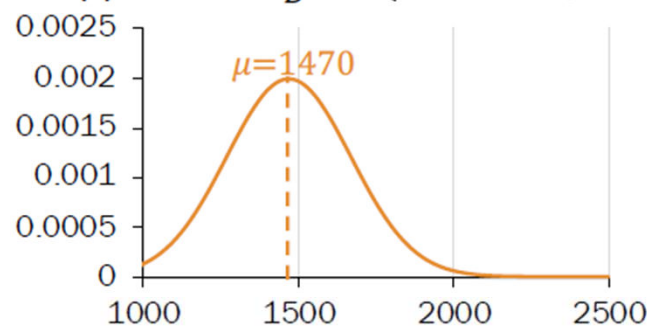
$$\begin{aligned} P(D > 0) &= 1 - F_D(0) = 1 - \Phi\left(\frac{0 - 187}{282.842}\right) \\ &\approx 0.74574 \end{aligned}$$

Compare with **0.7488**, calculated by sampling!

Warriors $A_W \sim \mathcal{N}(S = 1657, 200^2)$



Opponents $A_B \sim \mathcal{N}(S = 1470, 200^2)$



```
>>> from scipy.stats import norm
>>> 1 - norm(187, 80000 ** 0.5).cdf(0)
0.7457402843526317
>>> 1 - norm(0, 1).cdf(-187 / (80000 ** 0.5))
0.7457402843526317
```

Virus infections

Suppose you are working with the WHO to initiate a response to the onset of a virus. There are two exposed groups:

- G1: 20000 people, each independently infected with $p_1 = 0.1$
- G2: 10000 people, each independently infected with $p_2 = 0.4$

What is $P(\text{people infected} \geq 6100)$? An approximation is okay.

1. Define RVs & state goal

Let $A = \#$ infected in G1.
 $A \sim \text{Bin}(20000, 0.1)$
 $B = \#$ infected in G2.
 $B \sim \text{Bin}(10000, 0.4)$

Want: $P(A + B \geq 6100)$

Strategy:

A. Sum of independent Binomials

B. Sum of independent Poissons

C. Sum of independent Gaussians

D. Sum of independent Exponentials

no, $p_1 \neq p_2$
no, because p_1 and p_2 too large

yes!!!!

no, because mean and stdev of B are very different

Virus infections

Suppose you are working with the WHO to initiate a response to the onset of a virus. There are two exposed groups:

- G1: 20000 people, each independently infected with $p_1 = 0.1$
- G2: 10000 people, each independently infected with $p_2 = 0.4$

What is $P(\text{people infected} \geq 6100)$? An approximation is okay.

1. Define RVs
& state goal

Let $A = \#$ infected in G1.
 $A \sim \text{Bin}(20000, 0.1)$
 $B = \#$ infected in G2.
 $B \sim \text{Bin}(10000, 0.4)$

2. Approximate as sum of Gaussians

$$A \approx X \sim \mathcal{N}(2000, 1800) \quad B \approx Y \sim \mathcal{N}(4000, 2400)$$

$$P(A + B \geq 6100) \approx P(X + Y \geq 6099.5) \quad \text{continuity correction}$$

3. Solve

Want: $P(A + B \geq 6100)$

Virus infections

```
>>> 1 - norm(6000, 4200 ** 0.5).cdf(6099.5)
0.06235282662988528
>>> 1 - norm(0, 1).cdf((6099.5 - 6000)/(4200 ** 0.5))
0.06235282662988528
```

Suppose you are working with the WHO to initiate a response to the onset of a virus. There are two exposed groups:

- G1: 20000 people, each independently infected with $p_1 = 0.1$
- G2: 10000 people, each independently infected with $p_2 = 0.4$

What is $P(\text{people infected} \geq 6100)$? An approximation is okay.

1. Define RVs
& state goal

Let $A = \#$ infected in G1.
 $A \sim \text{Bin}(20000, 0.1)$
 $B = \#$ infected in G2.
 $B \sim \text{Bin}(10000, 0.4)$

Want: $P(A + B \geq 6100)$

2. Approximate as sum of Gaussians

$A \approx X \sim \mathcal{N}(2000, 1800)$ $B \approx Y \sim \mathcal{N}(4000, 2400)$
 $P(A + B \geq 6100) \approx P(X + Y \geq 6099.5)$ continuity correction

3. Solve

Let $W = X + Y \sim \mathcal{N}(6000, 4200)$
 $P(W \geq 6099.5) = 1 - \Phi\left(\frac{6099.5 - 6000}{\sqrt{4200}}\right)$
 $\approx 1 - \Phi(1.53531) \approx 0.06235$

Sum of independent Gaussians

$$\begin{array}{l} X_1 \sim \mathcal{N}(\mu_1, \sigma_1^2), \\ X_2 \sim \mathcal{N}(\mu_2, \sigma_2^2) \\ X_1, X_2 \text{ independent} \end{array} \quad \Rightarrow \quad X_1 + X_2 \sim \mathcal{N}(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$$

Is this related to linear transformations of Gaussians?

Recall:

$$\text{If } Y = \underline{a}X + \underline{b}, \text{ then } Y \sim \mathcal{N}(\underline{a}\mu_X + \underline{b}, \underline{a}^2\sigma_X^2)$$

Linear transforms vs. independence



Let $X \sim \mathcal{N}(\mu, \sigma^2)$ and $Y = X + X$. What is the distribution of Y ?

- Are both approaches valid?

Independent RVs approach



Let $X_1 \sim \mathcal{N}(\mu_1, \sigma_1^2)$, $X_2 \sim \mathcal{N}(\mu_2, \sigma_2^2)$
be independent.

Then $Y = X_1 + X_2 \sim \mathcal{N}(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$

$$Y = X + X$$

$$X + X \sim \mathcal{N}(\mu + \mu, \sigma^2 + \sigma^2)?$$

$$Y \sim \mathcal{N}(\underline{2\mu}, 2\sigma^2)?$$

X is NOT
independent
of X !

Linear transform approach



Let $X \sim \mathcal{N}(\mu, \sigma^2)$.

If $Y = aX + b$,
then $Y \sim \mathcal{N}(a\mu + b, a^2\sigma^2)$.

$$Y = 2X$$

$$Y \sim \mathcal{N}(\underline{2\mu}, 4\sigma^2)$$

For independent $X_1 \sim \mathcal{N}(\mu_1, \sigma_1^2)$, $X_2 \sim \mathcal{N}(\mu_2, \sigma_2^2)$,
 $aX_1 + bX_2 + c \sim \mathcal{N}(a\mu_1 + b\mu_2 + c, a^2\sigma_1^2 + b^2\sigma_2^2)$